Basic Concepts
Computational Thinking

Introduction to Computational Thinking Course WI1102 teaching team.

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Objectives

- Students are able to explain the basic concepts of computational thinking for problem solving: problem decomposition, abstraction, pattern recognition, and algorithmic thinking.
- Students are able to use the basic concepts of problem decomposition, abstraction, and pattern recognition to identify and explain a problem.

Computing is Everywhere



Mention any domain where computing is **not** useful.

Computing in Religion Study

Search engine on sacred texts, multimedia for learning sacred texts, find chapters and verses related to daily concepts, classification of prophetic narrations, etc.

Computing in History Study

Information retrieval on historical documents, digitizing hand-written historical documents, augmented and virtual reality tour at historical places, etc.

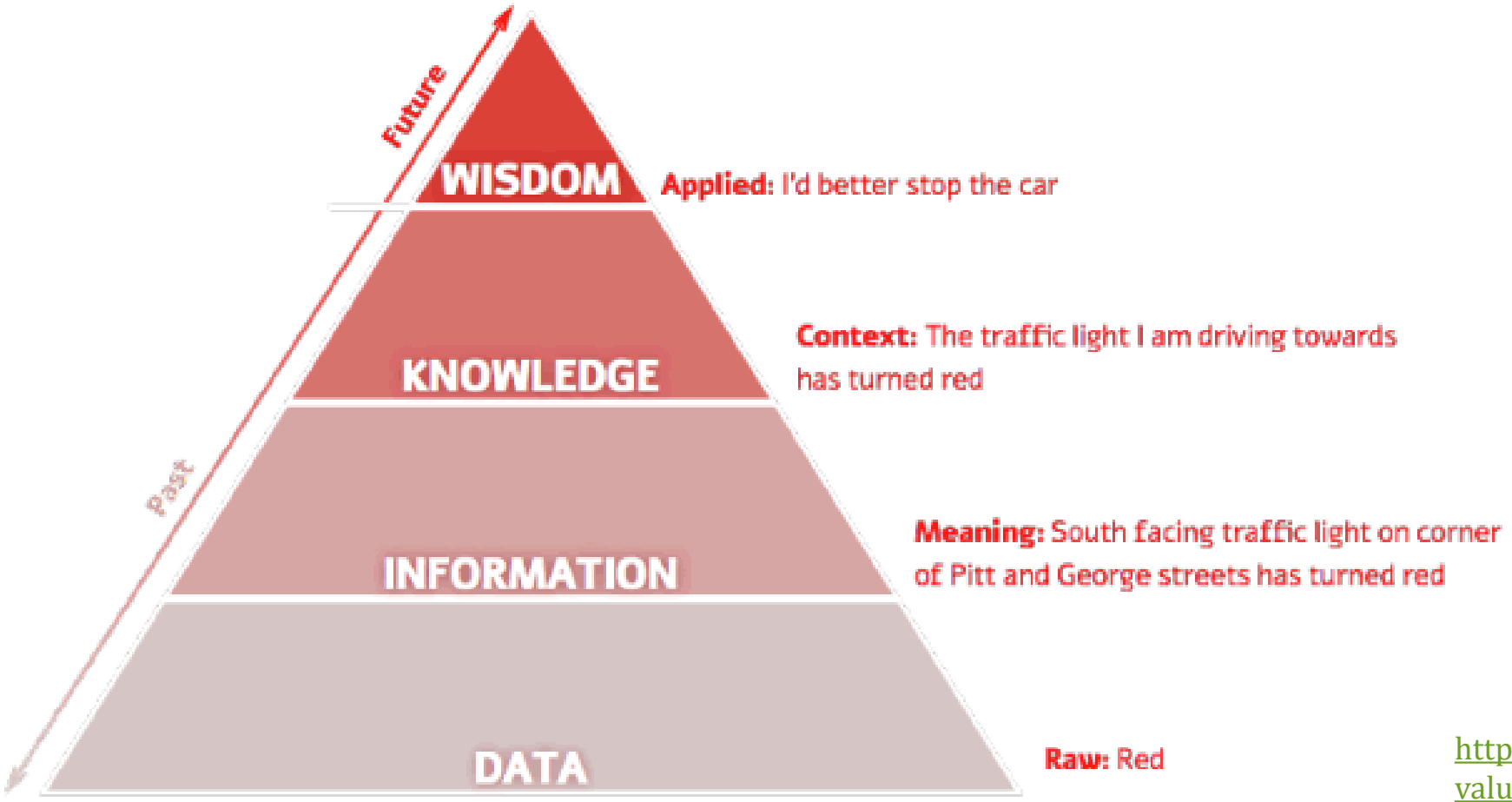
Computing in Military and Intelligence

Encryption for securing confidential communication, military inventory management, computer simulation for military training, object recognition, open source intelligence, etc.

Computing in Politics

Vote counting system, decision support systems, digital campaign, e-government services, etc.

Data – Information – Knowledge – Wisdom



https://www.i-scoop.eu/big-data-action-value-context/dikw-model/

What is Computational Thinking

Computational thinking is the thought processes involved in formulating a problem and expressing its solution(s) in such a way that a computer —human or machine— can effectively carry out.

"Human computers"??

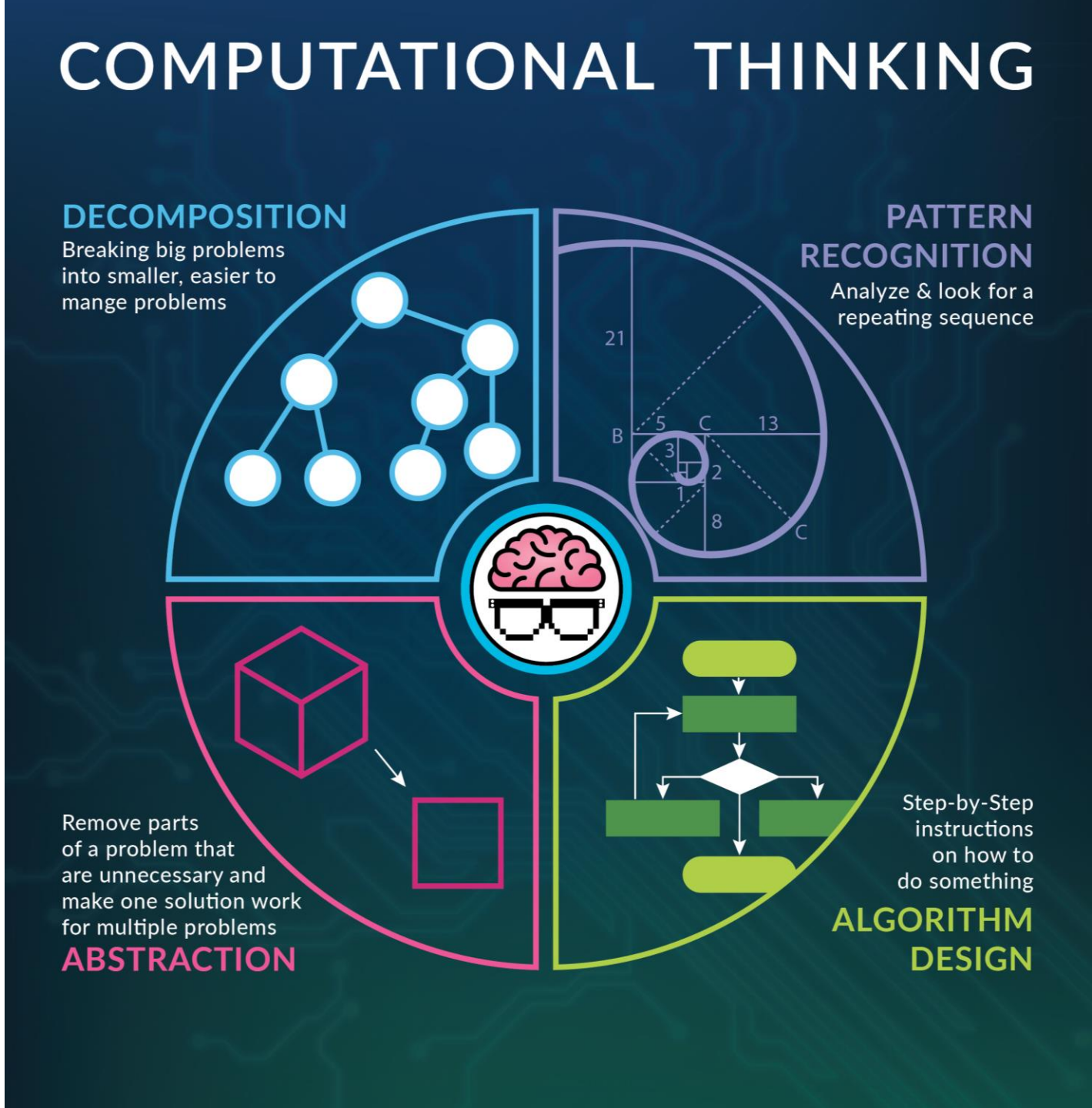
- Humans can compute
- People can learn computational thinking without a machine

http://socialissues.cs.toronto.edu/index.html%3Fp=279.html

Why Do I Need Computational Thinking?

- Computational thinking: Digital age skills for everyone
  - http://youtu.be/VFcUgSYyRPg

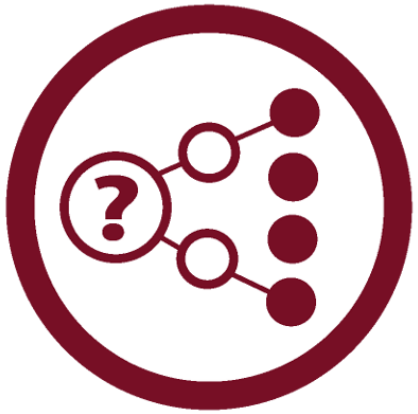
4 Pillars of Computational Thinking



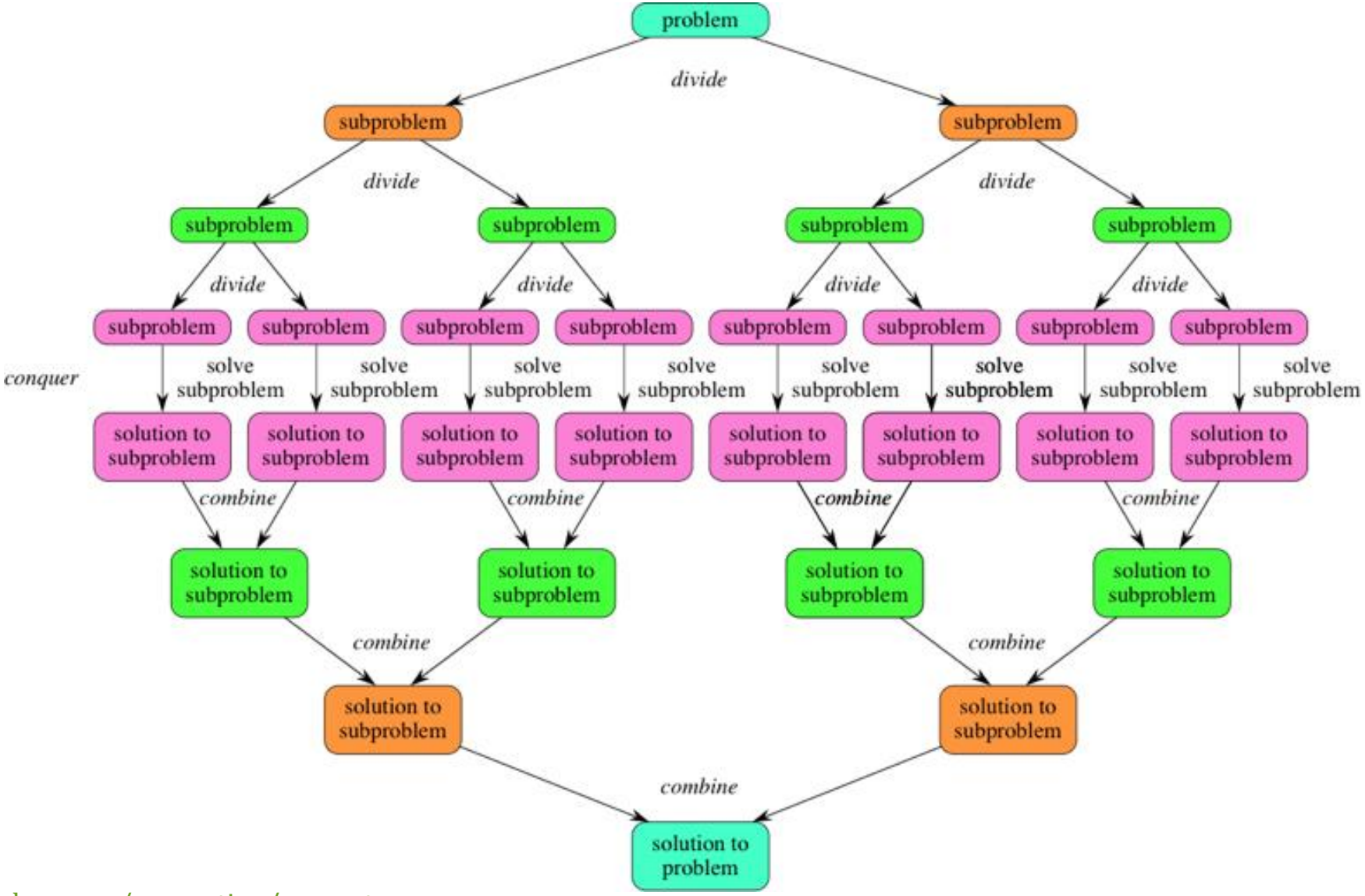
https://www.computationalthinkers.com/product/computationalthinking/

Problem Decomposition

Solving a problem by dividing the problem into smaller sub-problems, then solving these smaller problems separately.



Divide and Conquer Strategy



https://www.khanacademy.org/computing/computer-science/algorithms/merge-sort/a/divide-and-conquer-algorithms

Example of Problem Decomposition (1)

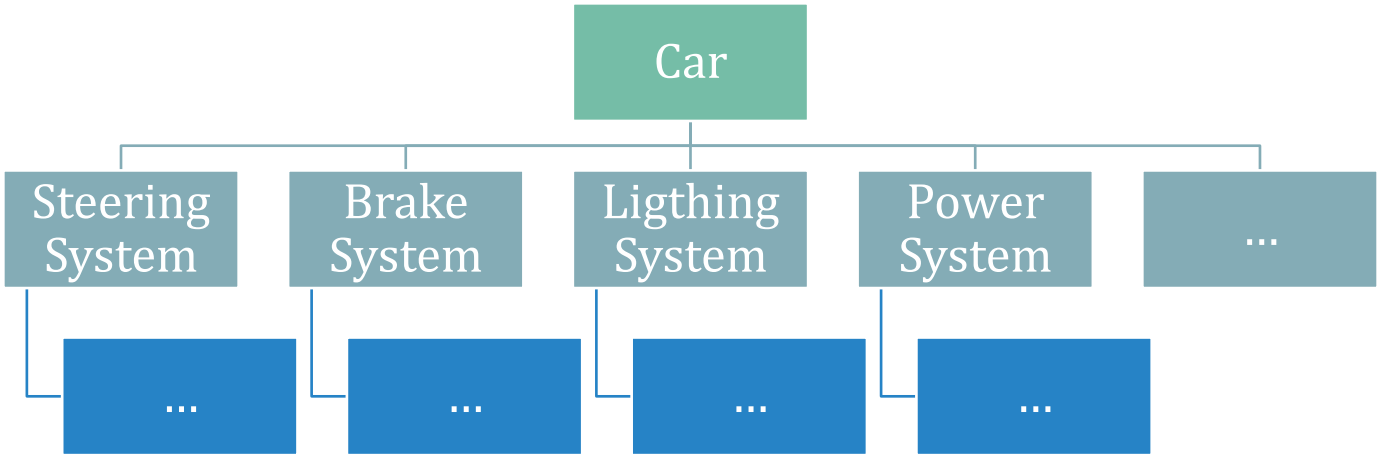
Decomposition of Scientific Article Writing

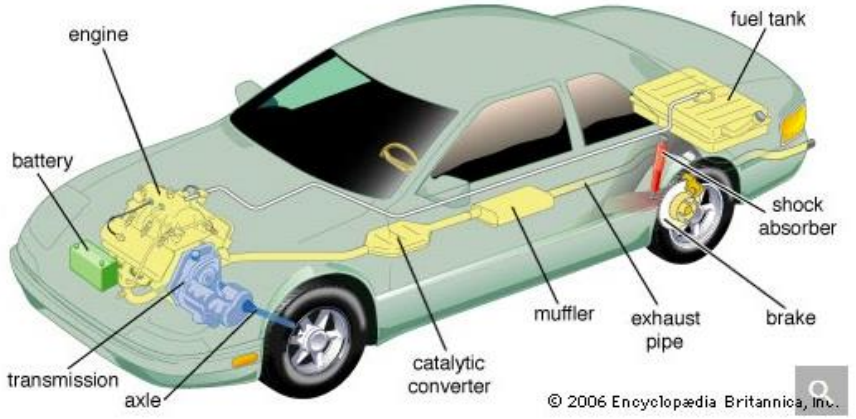
1. Main idea
    1. Sub-poin idea 1
        1. Sub poin 1.1
        2. Sub poin 1.1
    2. Sub-poin idea 1
        1. Sub poin 1.2
        2. Sub poin 1.2
            1. Sub poin 1.2.2
            2. …

2. Main Idea
    1. …

Example of Problem Decomposition (2)

Functional decomposition of a car, from the driver's perspective





The major functional components of an automobile.

Encyclopædia Britannica, Inc.

Example of Problem Decomposition

Classroom Decomposition

- Look at the classroom you are currently in.
- Note all the (inanimate) objects in the room.

Challenge:

Discuss the appropriate decomposition to describe the overall condition of the room.

Instructions:

- Decomposition can be done based on different aspects, for example: based on the types of objects present, based on their functions, etc.
- Decomposition can be done in stages.

Problem Decomposition Exercise

Academic Information System Decomposition

A brief description of an academic information system is given below:

Before the semester begins, all students are required to register for courses by completing a Study Plan Form (FRS), which must be approved by their academic advisor. Students may make changes to their Study Plan (PRS) no later than four weeks after the semester begins, which must also be approved by their academic advisor. At the beginning of each semester, students are also required to pay tuition fees through a designated bank account, which will be automatically recorded by the system. After tuition fees are paid, students can print their Student Study Card (KSM) as proof of course enrollment for that semester. Academic advisors, Heads of Study Programs, and administrative staff can also monitor the status of course enrollment and the status of KSM printing.

Challenge:

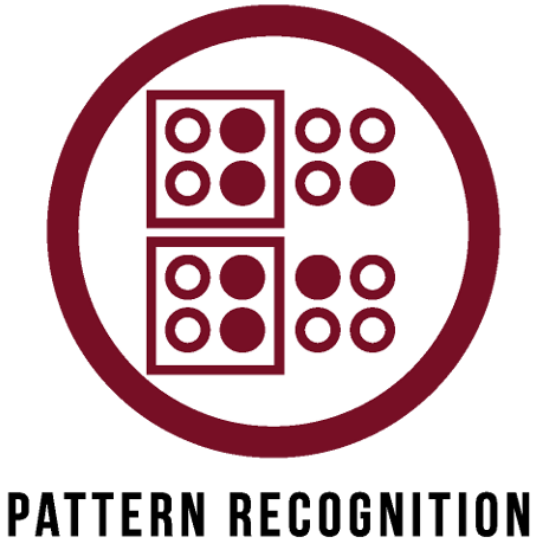
Discuss the decomposition of the modules (and their functions) that need to be developed to implement this academic information system.

Instructions:

Decomposition can be performed using various perspectives, for example, based on the type of user, the functions performed, the time at which the functions are executed, etc.

Pattern Recognition

Recognizing similarities/differences in the characteristics of various problems/phenomena that are different but related.



Pattern Recognition Example (1)

Find matching socks



Pattern Recognition Example (2)

Sequential Pattern Mining

Given a set of data containing a list of sequential events/activities.

For example, from:

- transaction records on an e-commerce platform
- patient medical records, etc.

We can obtain sequences of events that represent patterns believed to occur frequently, for example:

- Customers who purchase a smartphone will usually also purchase a screen protector and/or casing within the next three days.
- For a particular type of disease, what treatment patterns are commonly used and offer a better probability of recovery?

Pattern Recognition Exercise (1)

String Pattern

Some examples of international telephone numbers are given below:

+62-811-234567	+62 (817) 123-456
+1-888-300400	+1888300400
+44 (800) 987654	0065 7788 9900
0043-6543210	+966-200-300

Challenge:

Develop a list of rules that declare any string to be an international telephone number with the correct pattern.

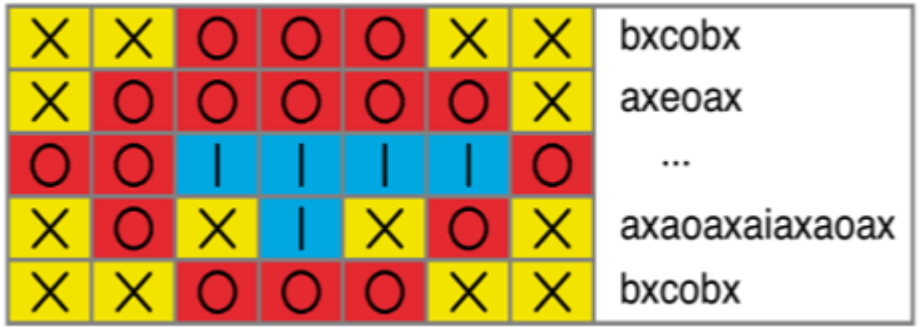
For example:

1. International telephone numbers can only contain numbers, +, -, (,), and spaces.
2. …

Pattern Recognition Exercise (2)

The Bebras Cipher

The colorful image was encoded by a program using the Bebras Cipher. The code can be seen on the right side of the following image, consisting of a string of letters. Unfortunately, the code for the third line is missing.



Challenge:

Which string of letters is the code for the missing third line?

Source: Bebras Indonesia (www.bebras.or.id). Kode Soal: I-2012-XXX-011-Sandi Bebras

Pattern Recognition Exercise (3)

Similarities in Characteristics

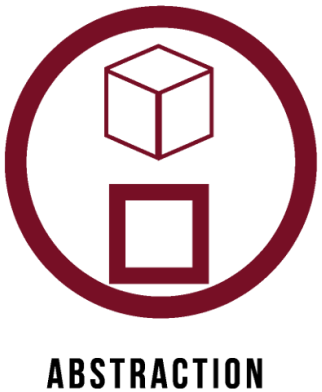
Discuss in groups with four friends around you:

Find seven characteristics that everyone in your group shares, except: age, gender, study program/school/faculty.



Abstraction and Generalization

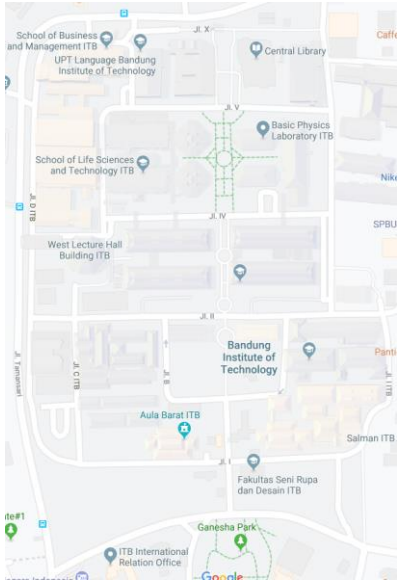
Eliminating unimportant parts of a problem (abstraction) to obtain a general solution to solving similar problems (generalization).



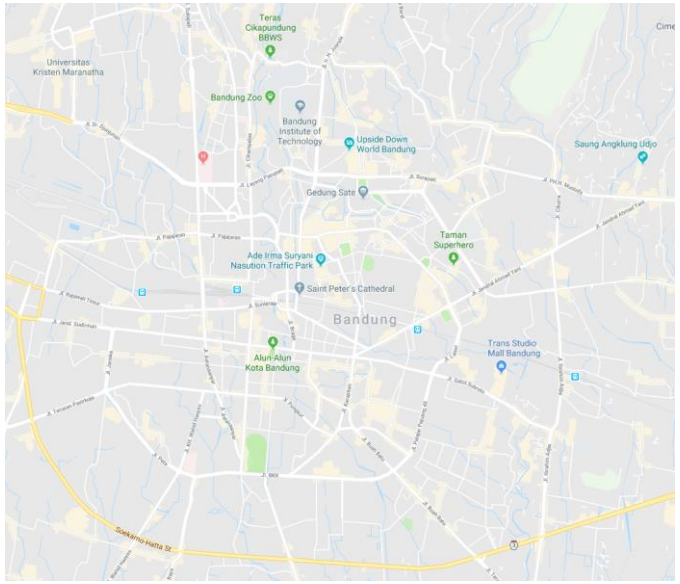
Abstraction Example

Abstraction Information

The information that appears on a map corresponds to the map level.



Map abstraction
at ITB level



Map abstraction
at Bandung level

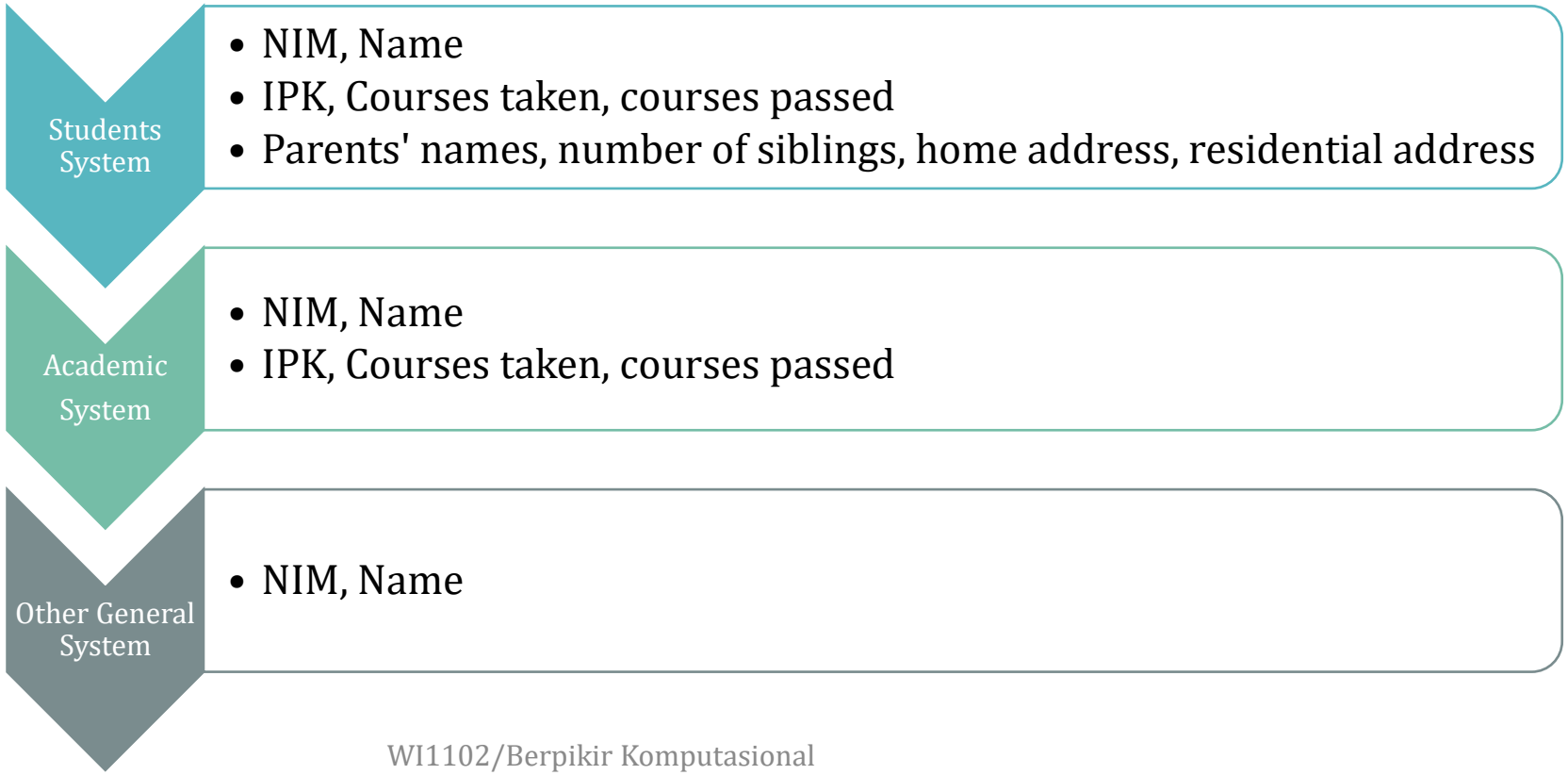


Map abstraction at Jawa Barat level
(What happens if all street names appear at this level?)

Example of data abstraction

Data abstraction of students

Depending on the scope of the information system used, student data can have different attribute details.

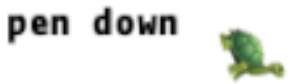


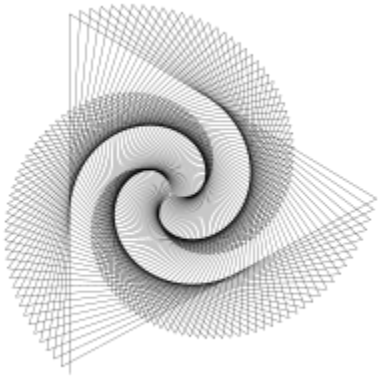
Abstraction Example - Generalization

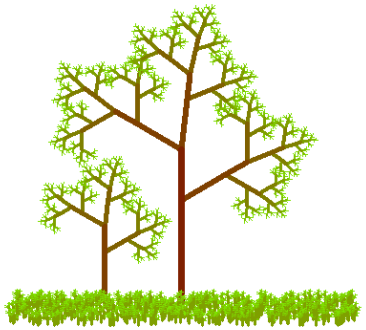
Turtle Graphics (Generalization Procedure)

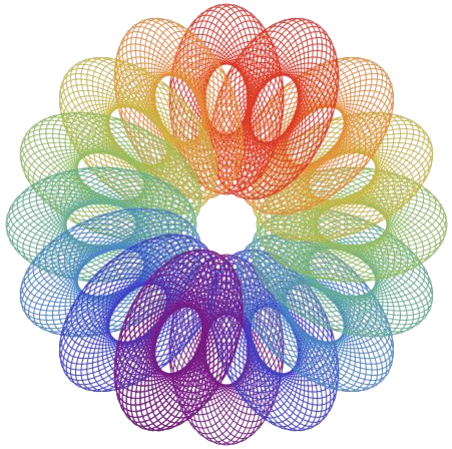
- The procedure for drawing arbitrary 2D vector graphics can be generalized using a turtle robot.
- The turtle robot has three attributes: position, direction, and pen. The turtle pen also has three attributes: on/off, color, and width.
- This turtle robot can receive three simple commands: move forward, rotate, and change pen attributes.
- Question: What attributes/commands must be added to the turtle robot to be able to draw arbitrary 3D graphics?

https://en.wikipedia.org/wiki/Turtle_graphics









Abstraction Exercise - Generalization

City Traveling Robot

A robot operates in a city. It travels along roads from one point to another, following instructions.

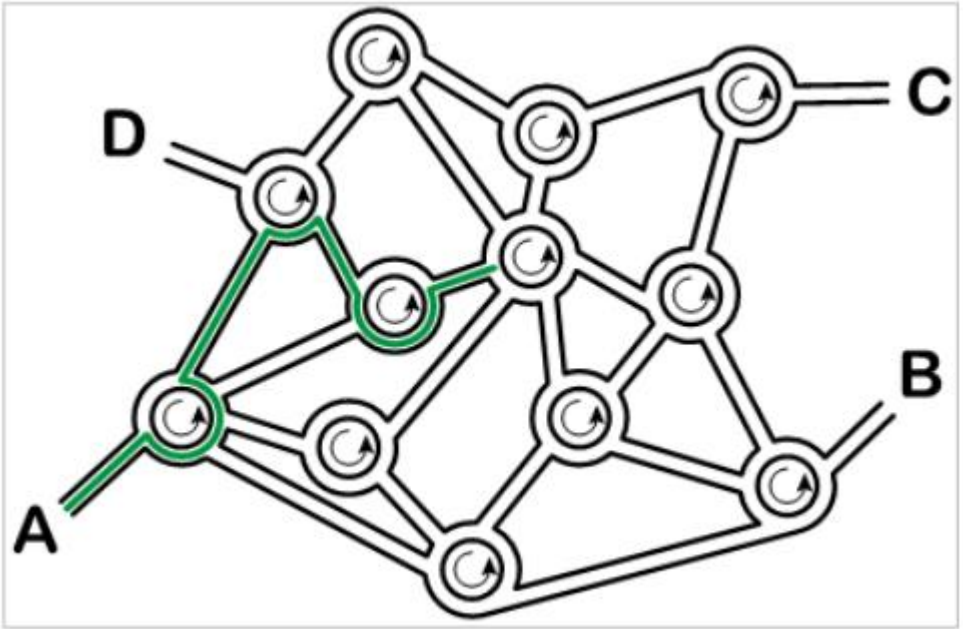
At each intersection, there are loops (detours), and the robot will turn counterclockwise, then take a specific sequence of detours according to the given instructions.

The instructions are in the form of numbers indicating the sequence of detours. For example, the instructions "4 1 2" will be followed by the robot as follows:

On the first turn, take the 4th detour.

On the second turn, take the 1st detour.

On the third detour, take the 2nd detour.



Challenge:

If the robot is initially at A, which instructions will lead it to C?

Source: Bebras Indonesia (www.bebras.or.id). Kode Soal: I-2017-Sl-02

Generalization Note:
Navigation from any point to any other point is simply done by providing a series of numbers!

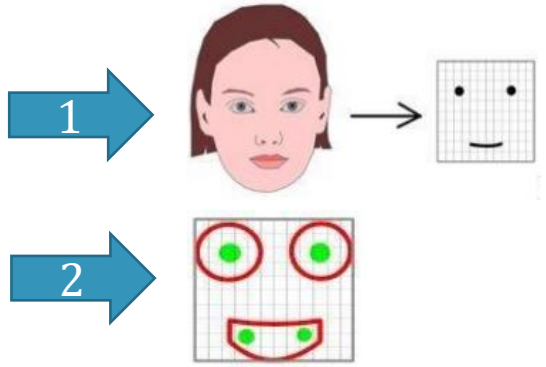
Generalization and Pattern Recognition Exercises

Smiling Face Recognition

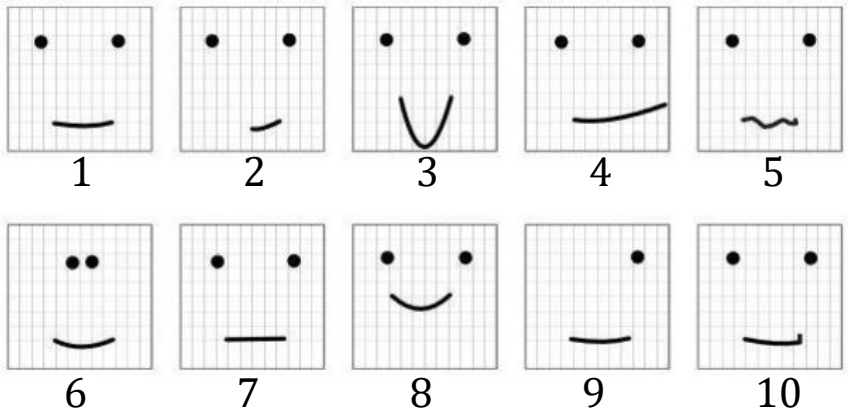
Beaver has invented a tool for detecting smiling faces using a camera. The tool works in two steps:

Transform a photo of a face into a smiley face, with two dots and a line indicating the position of the eyes and mouth.

Detect whether the face is smiling by matching the image from step 1 with a pattern consisting of red lines and green dots. A face is detected as smiling if and only if the image touches all the green dots and none of the red lines.



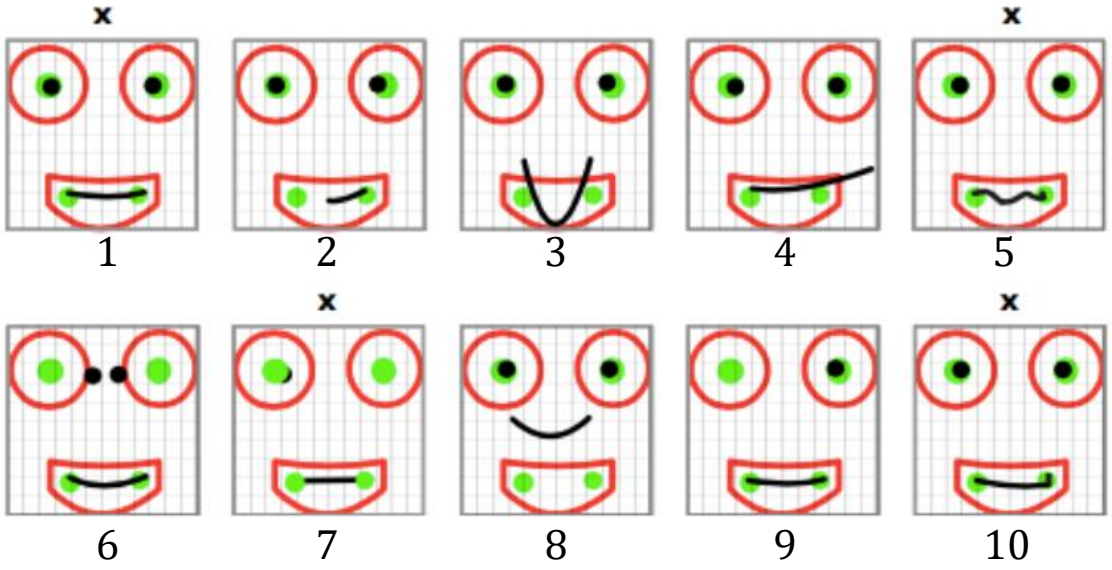
Challenge:
Which of the following images from the results of stage 1 are detected as smiling faces?



Source: Bebras Indonesia (www.bebras.or.id). Kode Soal: I-2017-DE-02

Generalization and Pattern Recognition Exercises

Smiling Face Recognition Answer



Images with an x mark qualify as smiley faces.

Moral of the story:

- Smiling face pattern recognition is simply based on the position of the eyes and the shape of the mouth (abstraction).
- A smiling face can be inferred using criteria based on the established red line boundaries (generalization).

Source: Bebras Indonesia (www.bebras.or.id). Jawaban Kode Soal: I-2017-DE-02

All-in-one Exercise

- Problem Statement: All WI1102 students must arrive in class and complete all homework/obligations before leaving.
- Decomposition (do individually):

Decompose the problem above into small steps that you typically take, from waking up to sitting in class.

- Pattern Recognition (discuss in groups of 3):

Look for similarities and differences in the small individual steps to create small steps for solving the problem together.

- Abstraction and Generalization (discuss in groups of 3 x 2):

- Combine the problem-solving steps from both groups.

- Abstraction: Remove steps that are deemed unnecessary (e.g., too specific or unimportant).

- Generalization: Formulate general problem-solving steps that can/need to be carried out by all WI1102 students to solve the problem.